



US012415877B2

(12) **United States Patent**
Jeon et al.

(10) **Patent No.:** **US 12,415,877 B2**

(45) **Date of Patent:** **Sep. 16, 2025**

(54) **ACRYLIC COPOLYMER COAGULANT AND METHOD OF PREPARING GRAFT COPOLYMER USING THE SAME**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 791 days.

(21) Appl. No.: **17/436,025**

(22) PCT Filed: **Jul. 15, 2020**

(86) PCT No.: **PCT/KR2020/009324**

§ 371 (c)(1),

(2) Date: **Sep. 2, 2021**

(87) PCT Pub. No.: **WO2021/015485**

PCT Pub. Date: **Jan. 28, 2021**

(65) **Prior Publication Data**

US 2022/0177621 A1 Jun. 9, 2022

(30) **Foreign Application Priority Data**

Jul. 24, 2019 (KR) 10-2019-0089774

(51) **Int. Cl.**

C08F 220/06 (2006.01)

C08C 1/15 (2006.01)

C08L 51/04 (2006.01)

(52) **U.S. Cl.**

CPC **C08F 220/06** (2013.01); **C08C 1/15** (2013.01); **C08L 51/04** (2013.01)

(58) **Field of Classification Search**

CPC C08F 220/06; C08F 220/14; C08F 220/1802; C08F 6/22; C08F 279/04; C08F 6/18; C08C 1/15; C08C 1/065; C08L 51/04; C08L 2207/04

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,487,890 A * 12/1984 Kishida C08F 279/02 525/193

4,694,032 A * 9/1987 Kakimoto C08L 51/04 524/217

5,204,406 A 4/1993 Fujii et al.

5,708,082 A 1/1998 Huang

2010/0048816 A1 2/2010 Ryu et al.

2011/0077355 A1 3/2011 Shikisai et al.

2016/0304652 A1 10/2016 Niessner et al.

FOREIGN PATENT DOCUMENTS

CN 109517286 A 3/2019

EP 0077038 A2 4/1983

JP H0859704 A * 3/1996

JP H10158460 A 6/1998

JP 2000-007874 A 1/2000

JP 3358885 B2 12/2002

JP 2003105161 A 4/2003

JP 3648861 B2 5/2005

JP 2006143924 A 6/2006

KR 950026896 A 10/1995

KR 1998-0057234 A 9/1998

KR 20100044215 A 4/2010

KR 101101092 B1 12/2011

KR 20180047749 A 5/2018

WO 2015-086164 A1 6/2015

OTHER PUBLICATIONS

English Translation of JPH0859704A (Year: 1996).*

Supplementary European Search Report for related EP application No. 20843694.9, mailed Sep. 6, 2022.

* cited by examiner

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(57) **ABSTRACT**

The present invention provides an acrylic copolymer coagulant which is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and an acrylic monomer, and a method of preparing a graft copolymer using the same. According to the present invention, a graft copolymer excellent in impact strength, surface gloss characteristics, and processability can be prepared.

6 Claims, No Drawings

1

ACRYLIC COPOLYMER COAGULANT AND METHOD OF PREPARING GRAFT COPOLYMER USING THE SAME

CROSS-REFERENCE TO RELATED APPLICATION

This application is a National Phase of International Application No. PCT/KR2020/009324, which claims priority to and the benefit of Korean Patent Application No. 10-2019-0089774, filed on Jul. 24, 2019, the disclosures of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

The present invention relates to an acrylic copolymer coagulant and a method of preparing a graft copolymer using the same, and particularly, to an acrylic copolymer coagulant whose composition and average particle diameter have been adjusted within specific ranges and a method of preparing a graft copolymer using the same.

BACKGROUND ART

Acrylonitrile-butadiene-styrene copolymers (hereinafter, referred to as "ABS copolymers") are formed by graft copolymerization of a butadiene rubber polymer with styrene and acrylonitrile.

ABS copolymers exhibit excellent physical properties such as high impact resistance, high chemical resistance, high thermal stability, high colorability, high fatigue resistance, high rigidity, high processability, and the like compared to existing high-impact polystyrene (HIPS), and, among them, especially, processability is excellent. Due to these characteristics, ABS copolymers can be used in interior and exterior materials for automobiles, office equipment, parts of various electric and electronic products, toys, or the like.

Meanwhile, in order to form ABS copolymers having excellent impact resistance, the particle diameter of the butadiene rubber polymer needs to be appropriately adjusted. Typically, when the average particle diameter thereof is in a range of 250 nm to 400 nm, excellent impact resistance may be realized without degradation of surface gloss characteristics. However, when a butadiene rubber polymer having the above-described average particle diameter is prepared by emulsion polymerization, a polymerization time is excessively prolonged, leading to degraded productivity. Accordingly, a method of preparing a butadiene rubber polymer having an average particle diameter of about 100 nm and then enlarging the butadiene rubber polymer with a coagulant has been proposed. However, when acetic acid is used as the coagulant in the enlargement, an excessive amount of aggregates is generated, and when the concentration of a butadiene rubber polymer latex is lowered to reduce the generation of aggregates, productivity is degraded. In order to solve these problems, a method of using an acrylic copolymer coagulant which is a copolymer of a carboxylic acid-based monomer and an acrylic monomer has been proposed, but this method also has a problem in which the impact strength and surface gloss characteristics of an ABS graft copolymer are degraded.

2

DISCLOSURE

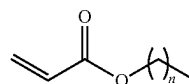
Technical Problem

The present invention provides an acrylic copolymer coagulant capable of improving the impact strength, surface gloss characteristics, processability, and latex stability of a graft copolymer.

Technical Solution

One aspect of the present invention provides an acrylic copolymer coagulant which is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and an acrylic monomer represented by Chemical Formula 1 below, wherein a content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant satisfies Expression 1 below, and an average particle diameter Y of the acrylic copolymer coagulant satisfies Expression 2 below:

<Chemical Formula 1>



In Chemical Formula 1,
n is 0 or 1.

$$2+(n+1)^2-0.5(n+1) \leq X \leq 2+(n+1)^2+0.5(n+1) \quad \text{<Expression 1>}$$

$$\frac{\{20.817 \times (n+1)^2 - 27.35 \times (n+1) + 28.433\} / X \times 10 - 1.5 \leq Y \leq \{20.817 \times (n+1)^2 - 27.35 \times (n+1) + 28.433\} / X \times 10 + 1.5}{\quad \text{<Expression 2>}}$$

In Expressions 1 and 2,
n is an n value of Chemical Formula 1,
X is a content of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant (units: wt %), and
Y is an average particle diameter of the acrylic copolymer coagulant (units: nm).

Another aspect of the present invention provides a method of preparing a graft copolymer, which comprises: preparing a second diene-based rubber polymer by coagulation of a first diene-based rubber polymer with the above-described acrylic copolymer coagulant; and preparing a graft copolymer by graft polymerization of the second diene-based rubber polymer with an aromatic vinyl-based monomer and a vinyl cyanide-based monomer.

Advantageous Effects

An acrylic copolymer coagulant according to the present invention is formed so that the content of a carboxylic acid-based monomer unit and the average particle diameter of the acrylic copolymer coagulant satisfy specific ranges according to the number of carbon atoms of an acrylic monomer which is a raw material. When such an acrylic copolymer coagulant of the present invention is used, a graft copolymer excellent in impact strength, surface gloss characteristics, processability, and appearance characteristics can be prepared, and the latex stability of the prepared graft copolymer can also be improved.

MODES OF THE INVENTION

Hereinafter, the present invention will be described in more detail to facilitate understanding of the present invention.

3

Terms and words used in this specification and claims should not be interpreted as being limited to commonly used meanings or meanings in dictionaries, and, based on the principle that the inventors can appropriately define concepts of terms in order to describe their invention in the best way, the terms and words should be interpreted with meanings and concepts which are consistent with the technological spirit of the present invention.

In the present invention, the average particle diameters of an acrylic copolymer coagulant and first and second diene-based rubber polymers may be measured by a dynamic light scattering method, specifically, by using Nicomp 380 HPL (manufactured by Nicomp). In the specification, an average particle diameter may refer to an arithmetic average particle diameter in the particle size distribution as measured by a dynamic light scattering method, specifically, an average particle diameter measured in the scattering intensity distribution.

In the present invention, a carboxylic acid-based monomer may be one or more selected from the group consisting of acrylic acid, methacrylic acid, itaconic acid, crotonic acid, fumaric acid, and maleic acid, with methacrylic acid being preferred.

In the present invention, a carboxylic acid-based monomer unit may refer to a unit derived from the carboxylic acid-based monomer.

In the present invention, the weight-average molecular weight of an acrylic copolymer coagulant may be measured by gel permeation chromatography (GPC) analysis after an acrylic copolymer coagulant latex is dissolved in tetrahydrofuran (THF) at a concentration of 2 mg/ml, stirred for 12 hours, and filtered through a 45-nm polytetrafluoroethylene filter.

In the present invention, a first diene-based rubber polymer is formed of a diene-based monomer, and the diene-based monomer may be one or more selected from the group consisting of 1,3-butadiene, isoprene, chloroprene, and piperylene, with 1,3-butadiene being preferred.

In the present invention, an aromatic vinyl-based monomer may be one or more selected from the group consisting of styrene, α -methylstyrene, α -ethylstyrene, and p-methylstyrene, with styrene being preferred.

In the present invention, an aromatic vinyl-based monomer unit may refer to a unit derived from the aromatic vinyl-based monomer.

In the present invention, a vinyl cyanide-based monomer may be one or more selected from the group consisting of acrylonitrile, 2-methylacrylonitrile, 3-phenyl-acrylonitrile, and α -chloroacrylonitrile, with acrylonitrile being preferred.

In the present invention, a vinyl cyanide-based monomer unit may refer to a unit derived from the vinyl cyanide-based monomer.

Hereinafter, the present invention will be described in detail with reference to embodiments so that those skilled in the art can easily carry out the present invention. However, the present invention may be embodied in several different forms, and therefore, is not limited to embodiments described herein.

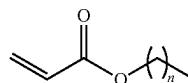
1. Acrylic Copolymer Coagulant

An acrylic copolymer coagulant according to an embodiment of the present invention is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and an acrylic monomer represented by Chemical Formula 1 below, wherein the content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant satisfies

4

Expression 1 below, and the average particle diameter Y of the acrylic copolymer coagulant satisfies Expression 2 below.

<Chemical Formula 1>



In Chemical Formula 1, n is 0 or 1.

$$2+(n+1)^2-0.5(n+1)\leq X\leq 2+(n+1)^2+0.5(n+1) \quad \text{<Expression 1>}$$

$$\frac{20.817\times(n+1)^2-27.35\times(n+1)+28.433}{X\times 10-1.5}\leq Y\leq \frac{20.817\times(n+1)^2-27.35\times(n+1)+28.433}{X\times 10+1.5} \quad \text{<Expression 2>}$$

In Expressions 1 and 2,

n is an n value of Chemical Formula 1,

X is a content of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant (units: wt %), and

Y is an average particle diameter of the acrylic copolymer coagulant (units: nm).

As described above, when an acrylic copolymer coagulant is used, the generation of aggregates can be reduced during coagulation of a diene-based rubber polymer, but there is a problem in which the impact strength and surface gloss characteristics of a graft copolymer are degraded. However, the inventors of the present invention have conducted research on an acrylic copolymer coagulant and found that when the acrylic copolymer coagulant is adjusted to have a specific composition and a specific average particle diameter, a graft copolymer excellent in impact strength, surface gloss characteristics, processability, and appearance characteristics can be prepared. Therefore, the present invention has been completed based on this fact.

Hereinafter, each component of the acrylic copolymer coagulant according to an embodiment of the present invention will be described in detail.

The acrylic copolymer coagulant is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and an acrylic monomer represented by Chemical Formula 1, wherein the content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant satisfies Expression 1, and the average particle diameter Y of the acrylic copolymer coagulant satisfies Expression 2.

When the acrylic copolymer coagulant satisfies all of Chemical Formula 1, Expression 1, and Expression 2, a graft copolymer excellent in impact strength, surface gloss characteristics, processability, and appearance characteristics can be prepared.

However, when the monomer mixture comprises an acrylic monomer not represented by Chemical Formula 1, it is necessary to use an excessive amount of a carboxylic acid-based monomer having the characteristics of coagulating other substances, and thus foreign matter such as gels and the like are excessively generated in a graft copolymer, leading to degraded appearance characteristics. In addition, the surface gloss characteristics of a graft copolymer may be substantially degraded.

When the content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant is below the range of Expression 1, even when the usage amount of the acrylic copolymer coagulant increases, it is not possible to enlarge a first diene-based rubber polymer so as to have a desired average particle diameter. As a result, the impact strength

5

and surface gloss characteristics of a graft copolymer may be degraded. Also, since an amount of aggregates increases in preparation of a graft copolymer, a process of removing the aggregates is additionally required, and thus production efficiency may be degraded. When the content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant is above the range of Expression 1, an aggregation phenomenon occurs during enlargement of a first diene-based rubber polymer, and thus it is not possible to prepare a graft copolymer.

When the average particle diameter Y of the acrylic copolymer coagulant is below or above the range of Expression 2, the impact strength and surface gloss characteristics of a graft copolymer may be substantially degraded.

Meanwhile, when the acrylic copolymer coagulant is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and methyl acrylate, the content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant may be in a range of 2.5 wt % to 3.5 wt %. When the content X is below the above-described range, even when the usage amount of the acrylic copolymer coagulant increases, it is not possible to enlarge a first diene-based rubber polymer so as to have a desired average particle diameter. As a result, the impact strength and surface gloss characteristics of a graft copolymer may be degraded. Also, since an amount of aggregates increases in preparation of a graft copolymer, a process of removing the aggregates is additionally required, and thus production efficiency may be degraded. When the content X is above the above-described range, an aggregation phenomenon occurs during enlargement of a first diene-based rubber polymer, and thus it is not possible to prepare a graft copolymer.

When the acrylic copolymer coagulant is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and methyl acrylate, and the content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant is in a range of 2.5 wt % to 3.5 wt %, the average particle diameter Y of the acrylic copolymer coagulant may be in a range of 80 nm to 115.5 nm. When the average particle diameter Y does not satisfy the above-described range, the impact strength and surface gloss characteristics of a graft copolymer may be substantially degraded.

Specifically, when the acrylic copolymer coagulant is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and methyl acrylate, and the content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant is 2.5 wt %, the average particle diameter of the acrylic copolymer coagulant may be in a range of 86.1 nm to 89.1 nm. In addition, when the acrylic copolymer coagulant is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and methyl acrylate, and the content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant is 3 wt %, the average particle diameter of the acrylic copolymer coagulant may be in a range of 71.5 nm to 74.5 nm. Additionally, when the acrylic copolymer coagulant is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and methyl acrylate, and the content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant is 3.5 wt %, the average particle diameter of the acrylic copolymer coagulant may be in a range of 61.5 nm to 64 nm.

When the acrylic copolymer coagulant is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and methyl acrylate, and the content X of a carboxylic acid-based monomer unit in the acrylic copoly-

6

mer coagulant is in a range of 2.5 wt % to 3.5 wt %, the acrylic copolymer coagulant may have a weight-average molecular weight of 700,000 g/mol to 950,000 g/mol or 700,000 g/mol to 930,000 g/mol. When the above-described range is satisfied, a graft copolymer with excellent processability can be prepared.

Meanwhile, when the acrylic copolymer coagulant is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and ethyl acrylate, the content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant may be in a range of 5 wt % to 7 wt %. When the content X is below the above-described range, even when the usage amount of the acrylic copolymer coagulant increases, it is not possible to enlarge a first diene-based rubber polymer so as to have a desired average particle diameter. As a result, the impact strength and surface gloss characteristics of a graft copolymer may be degraded. Also, since an amount of aggregates increases in preparation of a graft copolymer, a process of removing the aggregates is additionally required, and thus production efficiency may be degraded. When the content X is above the above-described range, an aggregation phenomenon occurs during enlargement of a first diene-based rubber polymer, and thus it is not possible to prepare a graft copolymer.

In addition, when the acrylic copolymer coagulant is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and ethyl acrylate, and the content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant is in a range of 5 wt % to 7 wt %, the average particle diameter Y of the acrylic copolymer coagulant may be in a range of 80 nm to 115.5 nm. When the average particle diameter Y does not satisfy the above-described range, the impact strength and surface gloss characteristics of a graft copolymer may be substantially degraded.

Specifically, when the acrylic copolymer coagulant is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and ethyl acrylate, and the content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant is 5 wt %, the average particle diameter of the acrylic copolymer coagulant may be in a range of 112.5 nm to 115.5 nm. In addition, when the acrylic copolymer coagulant is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and ethyl acrylate, and the content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant is 5.5 wt %, the average particle diameter of the acrylic copolymer coagulant may be in a range of 102.5 nm to 105 nm. Additionally, when the acrylic copolymer coagulant is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and ethyl acrylate, and the content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant is 6 wt %, the average particle diameter of the acrylic copolymer coagulant may be in a range of 93 nm to 96.5 nm. In addition, when the acrylic copolymer coagulant is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and ethyl acrylate, and the content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant is 6.5 wt %, the average particle diameter of the acrylic copolymer coagulant may be in a range of 86.5 nm to 89 nm. Additionally, when the acrylic copolymer coagulant is a copolymer of a monomer mixture comprising ethyl acrylate and a carboxylic acid-based monomer, and the content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant is 7 wt %, the average particle diameter of the acrylic copolymer coagulant may be in a range of 80 nm to 82.5 nm.

When the acrylic copolymer coagulant is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and ethyl acrylate, and the content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant is in a range of 5 wt % to 7 wt %, the acrylic copolymer coagulant may have a weight-average molecular weight of 250,000 g/mol to 350,000 g/mol or 290,000 g/mol to 340,000 g/mol. When the above-described range is satisfied, a graft copolymer with excellent processability can be prepared.

Meanwhile, the upper limit and lower limit of Y in Expression 2 may be indicated to have significant figures up to one decimal place.

2. Preparation of Acrylic Copolymer Coagulant

The acrylic copolymer coagulant according to an embodiment of the present invention may be prepared by a method comprising: providing a polymerization solution containing a monomer mixture comprising a carboxylic acid-based monomer and an acrylic monomer represented by Chemical Formula 1 and a first emulsifier; and adding a second emulsifier to a reactor and then performing polymerization while continuously adding the polymerization solution.

The content of the carboxylic acid-based monomer in the monomer mixture may be the same as the content of the carboxylic acid-based monomer unit in the acrylic copolymer coagulant.

Meanwhile, the total amount of the first emulsifier and second emulsifier may be 0.2 to 0.6 parts by weight or 0.3 to 0.5 parts by weight with respect to 100 parts by weight of the monomer mixture. When the above-described range is satisfied, the average particle diameter of the acrylic copolymer coagulant may be easily adjusted to satisfy Expression 2. Also, a first diene-based rubber polymer may be easily enlarged to have a desired average particle diameter while the generation of aggregates is minimized during enlargement of the first diene-based rubber polymer.

When the acrylic copolymer coagulant is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and methyl acrylate, and the content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant is 2.5 wt %, the weight ratio of a first emulsifier and a second emulsifier may be 55.5:44.5 to 57.5:42.5. When the acrylic copolymer coagulant is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and methyl acrylate, and the content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant is 3 wt %, the weight ratio of a first emulsifier and a second emulsifier may be 47.75:52.25 to 52:48. When the acrylic copolymer coagulant is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and methyl acrylate, and the content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant is 3.5 wt %, the weight ratio of a first emulsifier and a second emulsifier may be 20:80 to 26.75:73.25. When the above-described range is satisfied, an acrylic copolymer coagulant having an average particle diameter satisfying Expression 2 can be prepared.

When the acrylic copolymer coagulant is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and ethyl acrylate, and the content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant is 5 wt %, the weight ratio of a first emulsifier and a second emulsifier may be 78:22 to 78.75:21.25. When the acrylic copolymer coagulant is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and

ethyl acrylate, and the content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant is 6 wt %, the weight ratio of a first emulsifier and a second emulsifier may be 20:80 to 26.75:73.25. When the acrylic copolymer coagulant is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and ethyl acrylate, and the content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant is 7 wt %, the weight ratio of a first emulsifier and a second emulsifier may be 78:22 to 78.25:21.75. When the above-described range is satisfied, an acrylic copolymer coagulant having an average particle diameter satisfying Expression 2 can be prepared.

Meanwhile, the first emulsifier and the second emulsifier may be one or more selected from the group consisting of sodium dicyclohexyl sulfosuccinate, sodium dihexyl sulfosuccinate, sodium di-2-ethylhexyl sulfosuccinate, potassium di-2-ethylhexyl sulfosuccinate, sodium dioctyl sulfosuccinate, sodium dodecyl benzene sulfate, sodium octadecyl sulfate, sodium oleyl sulfate, sodium dodecyl sulfate, potassium octadecyl sulfate, potassium rosinat, and sodium rosinat, with sodium dodecyl benzene sulfonate being preferred.

Meanwhile, the polymerization solution may further comprise an initiator and water.

The initiator may be one or more selected from the group consisting of sodium persulfate, potassium persulfate, ammonium persulfate, potassium superphosphate, hydrogen peroxide, t-butyl peroxide, cumene hydroperoxide, p-methane hydroperoxide, di-t-butyl peroxide, t-butyl cumyl peroxide, acetyl peroxide, isobutyl peroxide, octanoyl peroxide, dibenzoyl peroxide, 3,5,5-trimethyl hexanol peroxide, t-butyl peroxy isobutyrate, azobisisobutyronitrile, azobis-2,4-dimethylvaleronitrile, azobis cyclohexane carbonitrile, and azobis methyl isobutyrate, with potassium persulfate being preferred.

The initiator may be added in an amount of 0.02 to 2 parts by weight or 0.07 to 0.8 parts by weight with respect to 100 parts by weight of the monomer mixture, with the range of 0.07 to 0.8 parts by weight being preferred. When the above-described range is satisfied, the copolymer of the monomer mixture has an appropriate weight-average molecular weight, and thus a diene-based rubber polymer can be easily enlarged.

The water may be ion exchanged water.

Before the polymerization solution is added to the reactor, water may be additionally added.

3. Method of Preparing Graft Copolymer

A method of preparing a graft copolymer according to another embodiment of the present invention comprises: preparing a second diene-based rubber polymer by coagulation of a first diene-based rubber polymer with the acrylic copolymer coagulant according to an embodiment of the present invention; and preparing a graft copolymer by graft polymerization of the second diene-based rubber polymer with an aromatic vinyl-based monomer and a vinyl cyanide-based monomer.

The first diene-based rubber polymer may have an average particle diameter of 80 nm to 120 nm or 90 nm to 110 nm, with the range of 90 nm to 110 nm being preferred. When the above-described range is satisfied, a problem in which productivity is degraded due to a polymerization time excessively prolonged in preparation of the first diene-based rubber polymer can be solved. Also, when the acrylic

copolymer coagulant is used, the first diene-based rubber polymer can be easily enlarged to have a desired average particle diameter.

The second diene-based rubber polymer may have an average particle diameter of 250 nm to 400 nm or 280 nm to 350 nm, with the range of 280 nm to 350 nm being preferred. When the above-described range is satisfied, a graft copolymer excellent in impact strength and surface gloss characteristics can be prepared.

The acrylic copolymer coagulant may coagulate the first diene-based rubber polymer in an amount of 0.1 to 2 parts by weight or 0.5 to 1.5 parts by weight with respect to 60 parts by weight of the first diene-based rubber polymer, with the range of 0.5 to 1.5 parts by weight being preferred. When the above-described range is satisfied, the first diene-based rubber polymer can be easily coagulated, and an aggregation phenomenon cannot occur.

The acrylic copolymer coagulant and the first diene-based rubber polymer may be in a latex form, and the contents thereof are based on solid content.

In the step of preparing a graft copolymer, one or more selected from the group consisting of an initiator, a redox-based catalyst, and water may be further added.

The types of the initiator have been described above, and cumene hydroperoxide is preferred.

The initiator may be added in an amount of 0.01 to 0.5 parts by weight or 0.03 to 0.3 parts by weight with respect to 100 parts by weight of the sum of the first diene-based rubber polymer, the aromatic vinyl-based monomer, and the vinyl cyanide-based monomer, with the range of 0.03 to 0.3 parts by weight being preferred. When the above-described range is satisfied, emulsion polymerization can be facilitated, and the amount of the initiator remaining in the prepared graft copolymer can be minimized.

The redox-based catalyst may be one or more selected from the group consisting of sodium formaldehyde sulfoxylate, disodium dihydrogen ethylenediaminetetraacetate, ferrous sulfate, dextrose, sodium pyrophosphate, anhydrous sodium pyrophosphate, and sodium sulfate, with one or more selected from the group consisting of sodium formaldehyde sulfoxylate, disodium dihydrogen ethylenediaminetetraacetate, and ferrous sulfate being preferred.

The redox-based catalyst may be added in an amount of 0.01 to 0.5 parts by weight or 0.03 to 0.3 parts by weight with respect to 100 parts by weight of the sum of the first diene-based rubber polymer, the aromatic vinyl-based monomer, and the vinyl cyanide-based monomer, with the range of 0.03 to 0.3 parts by weight being preferred. When the above-described range is satisfied, polymerization may be easily initiated at relatively low temperatures.

The water may be ion exchanged water.

After the polymerization of the graft copolymer is completed, coagulation, aging, washing, drying processes may be further performed to obtain a graft copolymer in a powder form.

Hereinafter, exemplary embodiments of the present invention will be described in order to aid in understanding the present invention. However, it is apparent to those skilled in the art that the following embodiments are merely presented to exemplify the present invention, and various changes and modifications can be made to the exemplary embodiments of the present invention without departing from the scope and spirit of the present invention, so that the present invention covers all such changes and modifications provided they are within the scope of the appended claims.

EXAMPLES AND COMPARATIVE EXAMPLES

<Preparation of Acrylic Copolymer Coagulant>

100 parts by weight of a monomer mixture comprising methacrylic acid (MAA), methyl acrylate (MA), ethyl acrylate (EA), and butyl acrylate (BA) in contents shown in Tables 1 to 16 below, 0.4 parts by weight of potassium persulfate, 170 parts by weight of ion exchanged water, and sodium dodecyl benzene sulfonate as a first emulsifier in a content shown in Tables 1 to 16 below were uniformly mixed to prepare a polymerization solution.

170 parts by weight of ion exchanged water and sodium dodecyl benzene sulfonate as a second emulsifier in a content shown in Tables 1 to 16 below were added to a nitrogen-substituted polymerization reactor, and the temperature inside the reactor was raised to 80° C. Polymerization was performed while continuously adding the polymerization solution to the reactor at a predetermined rate for 5 hours to prepare an acrylic copolymer coagulant latex.

<Preparation of Second Butadiene-Based Rubber Polymer>

60 parts by weight (based on solid content) of a first butadiene rubber polymer latex having an average particle diameter of 103 nm was heated to 50° C. while stirring. Then, a coagulant was added in a content (based on solid content) shown in Tables 1 to 16 below, and aging was performed for 15 minutes. Afterward, 0.18 parts by weight of KOH was added, and aging was performed for 5 minutes while stirring to prepare a second butadiene rubber polymer latex.

<Preparation of Graft Copolymer>

The whole amount of the second butadiene rubber polymer latex, 3 parts by weight of styrene, 1 part by weight of acrylonitrile, and 93.9 parts by weight of ion exchanged water were added to a nitrogen-substituted reactor and then stirred. Subsequently, 0.12 parts by weight of cumene hydroperoxide and, as redox-based catalysts, 0.11 parts by weight of dextrose, 0.08 parts by weight of sodium pyrophosphate, and 0.0016 parts by weight of ferrous sulfate were batch-added. Then, the reactor was heated to 70° C. for 50 minutes. Subsequently, 27 parts by weight of styrene and 9 parts by weight of acrylonitrile were continuously added at a predetermined rate for 90 minutes, and then the reaction was terminated to prepare a graft copolymer latex.

The graft copolymer latex was subjected to coagulation with MgSO₄, aging, washing, dehydration, and drying to prepare a graft copolymer powder.

<Preparation of Thermoplastic Resin Composition>

25 parts by weight of the graft copolymer powder and 75 parts by weight of 92HR (commercially available from LG Chem Ltd., styrene/acrylonitrile copolymer) were mixed to prepare a thermoplastic resin composition.

Experimental Example 1

Properties of the acrylic copolymer coagulants of Examples and Comparative Examples were evaluated by methods described below, and results thereof are shown in Tables.

(1) Average particle diameter (nm): measured by a dynamic light scattering method using a Nicomp 380 HPL instrument (manufactured by Nicomp).

(2) Weight-average molecular weight (g/mol): measured by gel permeation chromatography (GPC) analysis after the acrylic copolymer coagulant latex was dissolved in tetrahydrofuran (THF) at a concentration of 2 mg/ml, stirred for 12 hours, and filtered through a 45-nm polytetrafluoroethylene filter.

11

(3) Polymerization conversion rate (%): determined by drying 1 g of the acrylic copolymer coagulant latex in an oven set at 135° C. for 20 minutes, measuring a solid content, and dividing the measured solid content by a theoretical solid content (target TSC).

Experimental Example 2

Physical properties of the second butadiene rubber polymer latex of Examples and Comparative Examples were measured by methods described below, and results thereof are shown in Tables.

(4) Aggregation phenomenon: When the measured amount (ppm) of aggregates is 30,000 ppm or more or when phase separation was visually observed during the stirring of the second butadiene rubber polymer latex, it was determined that an aggregation phenomenon occurred.

(5) Average particle diameter (nm): measured by a dynamic light scattering method using a Nicomp 380 HPL instrument (manufactured by Nicomp).

(6) Particle size distribution: measured by a dynamic light scattering method using a Nicomp 380 HPL instrument (manufactured by Nicomp).

Experimental Example 3

A property of graft copolymers of Examples and Comparative Examples was measured by a method described below, and results thereof are shown in Tables.

(7) Amount of aggregate (ppm): The graft copolymer latex was filtered through a 100-mesh sieve, placed in a convection oven, and allowed to stand at 80° C. for 720 minutes. Afterward, aggregates filtered on the 100-mesh sieve were weighed, and the amount of aggregates for the graft copolymer latex was calculated by the following equation.

12

Amount of aggregate (ppm)={Weight of aggregates filtered on 100-mesh sieve)/(Total theoretical weight of butadiene rubber polymer, styrene, acrylonitrile, and additives added in preparation of graft copolymer)}×1,000,000

Experimental Example 4

Each of the thermoplastic resin compositions of Examples and Comparative Examples was extruded to prepare a pellet. A property of the pellet was measured by a method described below, and results thereof are shown in Tables.

(8) Melt flow index (g/10 min): measured in accordance with ASTM D1238 at 220° C. and 10 kg.

Experimental Example 5

Each of the thermoplastic resin compositions of Examples and Comparative Examples was extruded and injection-molded to prepare a specimen. Properties of the specimen were measured by methods described below, and results thereof are shown in Table.

(9) Impact strength (kg m/m, ¼ In): measured in accordance with ASTM D256.

(10) Surface gloss characteristics: determined in accordance with ASTM D523 by measuring the injection-molded specimen at 45° using a gloss meter instrument (VG-7000+ Cu-2 manufactured by Nippon Denshoku Industries Co., Ltd.).

(11) Gel evaluation: The pellet was extruded into a film in a film extruder and simultaneously evaluated in real time using an optical gel counter, and the size and number per unit area of gels were measured and evaluated. Evaluation criteria are as follows.

○: Gel size of 100 μm or more and 100 or less gels

△: Gel size of 100 μm or more and 250 or less gels

x: Gel size of 100 μm or more and more than 250 gels

TABLE 1

Classification			Comparative Examples			
			1	2	3	4
Acrylic coagulant	Monomer mixture	MAA	2	2	2.5	2.5
		MA	98	98	97.5	97.5
		EA	—	—	—	—
		BA	—	—	—	—
	First emulsifier (parts by weight)		0.299	0.299	0.207	0.219
	Second emulsifier (parts by weight)		0.101	74.75	0.193	0.181
	n of Chemical Formula 1		0			
	X of Expression 1		2.5~3.5			
	Y of Expression 2		108~111	86.1~89.1		
	(1) Average particle diameter		110	100	75	85.5
Second diene-based rubber polymer	(2) Weight-average molecular weight		921,000	921,000	909,000	890,000
	(3) Polymerization conversion rate		99.7	99.9	99.9	99.8
	Addition amount of coagulant (parts by weight)		0.66	0.66	0.66	0.66
	(4) Aggregation phenomenon		x	x	x	x
	(5) Average particle diameter		198	220	284	288
	(6) Particle size distribution		0.68	0.59	0.64	0.66
	(7) Amount of aggregate		730	400	350	120

TABLE 1-continued

Classification		Comparative Examples			
		1	2	3	4
Thermoplastic resin composition	(8) Melt flow index	15	15.8	16.9	17.4
	(9) Impact strength	8.9	12.7	21.3	23.3
	(10) Surface gloss characteristics	92.3	89.6	90.9	97
	(11) Gel evaluation	○	Δ	○	○

TABLE 2

Classification			Examples			15
			1	2	3	
Acrylic copolymer coagulant	Monomer mixture	MAA	2.5	2.5	2.5	
		MA	97.5	97.5	97.5	
		EA	—	—	—	
		BA	—	—	—	
	First emulsifier (parts by weight)		0.222	0.225	0.23	20
		Second emulsifier (parts by weight)	0.178	0.175	0.17	
	n of Chemical Formula 1		0			25
	X of Expression 1		2.5~3.5			
	Y of Expression 2		86.1~89.1			30
	(1) Average particle diameter		86.5	88	89	
	(2) Weight-average molecular weight		912,000	918,000	909,000	
	(3) Polymerization conversion rate		99.7	99.9	99.9	

TABLE 2-continued

Classification		Examples		
		1	2	3
Second diene-based rubber polymer	Addition amount of coagulant (parts by weight)	0.66	0.66	0.66
	(4) Aggregation phenomenon	×	×	×
	(5) Average particle diameter	299	289	300
	(6) Particle size distribution	0.65	0.64	0.61
	(7) Amount of aggregate	120	95	200
	(8) Melt flow index	18.1	18	17.9
	(9) Impact strength	30.8	31	32.2
Thermoplastic resin composition	(10) Surface gloss characteristics	98.0	98.8	98.9
	(11) Gel evaluation	○	○	○

TABLE 3

Classification			Comparative Examples				
			5	6	7	8	9
Acrylic copolymer coagulant	Monomer mixture	MAA	2.5	2.5	3	3	3
		MA	97.5	97.5	97	97	97
		EA	—	—	—	—	—
		BA	—	—	—	—	—
	First emulsifier (parts by weight)		0.232	0.3	0.14	0.19	0.189
	Second emulsifier (parts by weight)		0.168	0.1	0.26	0.21	0.211
	n of Chemical Formula 1		0				
	X of Expression 1		2.5~3.5				
	Y of Expression 2		86.1~89.1			71.5~74.5	
	(1) Average particle diameter		89.5	100	67	70.5	71
Second diene-based rubber polymer	(2) Weight-average molecular weight		901,000	911,000	830,000	801,000	800,000
	(3) Polymerization conversion rate		99.7	99.9	99.9	100	99.9
	Addition amount of coagulant (parts by weight)		0.66	0.66	0.66	0.66	0.66
	(4) Aggregation phenomenon		×	×	×	×	×
	(5) Average particle diameter		290	297	300	299	297
	(6) Particle size distribution		0.66	0.66	0.54	0.54	0.59
	(7) Amount of aggregate		100	200	380	400	200
Thermoplastic resin composition	(8) Melt flow index		17.4	17.4	16.5	16	16.1
	(9) Impact strength		24	22	22.2	23	23.2
	(10) Surface gloss characteristics		97.2	96.2	97.2	97.3	97.3
	(11) Gel evaluation		?	?	?	?	?

15

TABLE 4

Classification			Examples			
			4	5	6	
Acrylic copolymer coagulant	Monomer mixture	MAA	3	3	3	5
		MA	97	97	97	
		EA	—	—	—	
		BA	—	—	—	
	First emulsifier (parts by weight)		0.191	0.2	0.208	10
	Second emulsifier (parts by weight)		0.209	0.2	0.192	
	n of Chemical Formula 1		0			
	X of Expression 1		2.5~3.5		15	
	Y of Expression 2		71.5~74.5			
	(1) Average particle diameter	71.5	73	74.5		
(2) Weight-average molecular weight	817,000	825,000	820,000			
Second diene-based rubber polymer	(3) Polymerization conversion rate	99.8	99.7	99.7	20	
	Addition amount of coagulant (parts by weight)	0.66	0.66	0.66		
	(4) Aggregation phenomenon	×	×	×		
	(5) Average particle diameter	301	296	310		25
	(6) Particle size distribution	0.62	0.63	0.6		
	(7) Amount of aggregate	200	100	100		
	(8) Melt flow index	17.9	17.7	18	30	
	(9) Impact strength	32.2	31.6	30.4		
	(10) Surface gloss characteristics	98.2	99	98.7		
	(11) Gel evaluation	?	?	?		35

TABLE 5

Classification			Comparative Examples				
			10	11	12	13	
Acrylic coagulant	Monomer mixture	MAA	3	3	3.5	3.5	40
		MA	97	97	96.5	96.5	
		EA	—	—	—	—	
		BA	—	—	—	—	
	First emulsifier (parts by weight)	0.21	0.25	0.05	0.056		
	Second emulsifier (parts by weight)	0.19	0.15	0.35	0.344		
	n of Chemical Formula 1	0					50
	X of Expression 1	2.5~3.5					
	Y of Expression 2	71.5~74.5	about 61.07~64.07				
	(1) Average particle diameter	75	85	59	60.5	55	
(2) Weight- average molecular weight	811,000	819,000	708,000	710,000			
(3) Polymerization conversion rate	99.9	99.9	99.9	>99.9			
Addition amount of coagulant (parts by weight)	0.66	0.66	0.66	0.66	60		
(4) Aggregation phenomenon	×	×	×	×			
(5) Average particle diameter	288	271	256	270			
(6) Particle size distribution	0.59	0.61	0.56	0.6			
(7) Amount of aggregate	210	390	450	290	65		

16

TABLE 5-continued

Classification		Comparative Examples			
		10	11	12	13
Thermo-plastic resin composition	(8) Melt flow index	14.7	17.2	15.5	16.2
	(9) Impact strength	23.7	19.9	8.9	16.6
	(10) Surface gloss characteristics	97	97.1	95.5	95.4
	(11) Gel evaluation	?	?	Δ	Δ

TABLE 6

Classification			Examples			
			7	8	9	10
Acrylic copolymer coagulant	Monomer mixture	MAA	3.5	3.5	3.5	3.5
		MA	96.5	96.5	96.5	96.5
		EA	—	—	—	—
		BA	—	—	—	—
	First emulsifier (parts by weight)		0.08	0.1	0.109	0.107
	Second emulsifier (parts by weight)		0.32	0.3	0.291	0.293
	n of Chemical Formula 1		0			
	X of Expression 1		2.5~3.5			
	Y of Expression 2		about 61.07~64.07			
	(1) Average particle diameter		61.5	63	63.5	64
Second diene-based rubber polymer	(2) Weight-average molecular weight		707,000	719,000	710,000	706,000
	(3) Polymerization conversion rate		99.8	100	99.9	>99.9
	Addition amount of coagulant (parts by weight)		0.66	0.66	0.66	0.55
	(4) Aggregation phenomenon		×	×	×	×
	(5) Average particle diameter		320	310	305	300
	(6) Particle size distribution		0.6	0.63	0.63	0.60
	(7) Amount of aggregate		100	93	120	190
	(8) Melt flow index		17.6	17.7	18	17.3
	(9) Impact strength		33.1	32.1	32.7	32.5
	(10) Surface gloss characteristics		98	98.1	99.1	98.2
Thermo-plastic resin composition	(11) Gel evaluation		?	?	?	?

17

TABLE 7

Classification			Comparative Examples			
			14	15	16	17
Acrylic coagulant	Monomer mixture	MAA	3.5	3.5	4	4
		MA	96.5	96.5	96	96
		EA	—	—	—	—
	First emulsifier (parts by weight)	BA	—	—	—	—
			0.109	0.2	0.064	0.040
			0.291	0.2	0.336	0.36
	Second emulsifier (parts by weight)					
		n of Chemical		1		
		Formula 1		2.5~3.5		
	Expression 1	Y of	about 61.07~64.07		53.25~56.25	
		Expression 2				
		(1) Average particle diameter	64.5	74	58	55
	(2) Weight-average molecular weight		740,000	711,000	699,000	687,000
		(3)	>99.9	>99.	>99.	99.9
	Polymerization conversion rate					

18

TABLE 7-continued

Classification			Comparative Examples			
			14	15	16	17
5	Addition amount of coagulant (parts by weight)		0.66	0.66	0.66	0.66
		(4) Aggregation phenomenon	×	×	?	?
		(5) Average particle diameter	309	333	—	—
	(6) Particle size distribution		0.60	0.63	—	—
		(7) Amount of aggregate	890	350	—	—
		(8) Melt flow index	15.7	16.6	—	—
	(9) Impact strength		26.0	20.6	—	—
		(10) Surface gloss	96.3	95.1	—	—
		characteristics				
	(11) Gel evaluation		Δ	Δ	—	—
10	Second diene-based rubber polymer					
	Thermo-plastic resin composition					
15	Thermo-plastic resin composition					
	Thermo-plastic resin composition					
20	Thermo-plastic resin composition					
	Thermo-plastic resin composition					
25	Thermo-plastic resin composition					
	Thermo-plastic resin composition					

TABLE 8

Classification			Comparative Examples			
			18	19	20	21
Acrylic coagulant	Monomer mixture	MAA	4	4.5	5	5
		MA	—	—	—	—
		EA	96	96	95	95
		BA	—	—	—	—
	First emulsifier (parts by weight)	0.336	0.34	0.297	0.31	
	Second emulsifier (parts by weight)	0.064	0.06	0.103	0.09	
	n of Chemical Formula 1	1	1	1	1	
	X of Expression 1	5~7	5~7	5~7	5~7	
	Y of Expression 2	about 141~444	about 125.17~128.17	about 112.5~115.5	about 112.5~115.5	
	(1) Average particle diameter	143	126	100	112	
	(2) Weight-average molecular weight	350,000	360,000	320,000	330,000	
Second diene-based rubber polymer	(3) Polymerization conversion rate	>99.9	99.8	99.9	>99.9	
		Addition amount of coagulant (parts by weight)	1.2	1.2	1.2	1.2
		(4) Aggregation phenomenon	×	×	×	×
	(5) Average particle diameter	191	201	278	311	
	(6) Particle size distribution	0.61	0.62	0.51	0.66	
	(7) Amount of aggregate	700	1,000	1280	200	
	(8) Melt flow index	14.8	15.0	17.2	16.4	
		(9) Impact strength	10.9	11.1	19.9	27.9
		(10) Surface gloss characteristics	88.3	87.9	85.9	98.7
		(11) Gel evaluation	×	×	×	?

19

TABLE 9

			Examples		
Classification			11	12	13
Acrylic copolymer coagulant	Monomer mixture	MAA	5	5	5
		MA	—	—	—
		EA	95	95	95
		BA	—	—	—
	First emulsifier (parts by weight)		0.312	0.313	0.315
	Second emulsifier (parts by weight)		0.088	0.087	0.085
	n of Chemical Formula 1		1		
	X of Expression 1		5~7		
	Y of Expression 2		about 112.5~115.5		
	(1) Average particle diameter		112.5	114	115.5
	(2) Weight-average molecular weight		330,000	322,000	334,000
	(3) Polymerization conversion rate		99.9	>99.9	99.9

20

TABLE 9-continued

			Examples		
Classification			11	12	13
Second diene-based rubber polymer	Addition amount of coagulant (parts by weight)		1.2	1.2	1.2
	(4) Aggregation phenomenon		×	×	×
	(5) Average particle diameter		299	293	302
	(6) Particle size distribution		0.55	0.55	0.56
	(7) Amount of aggregate		150	120	100
	(8) Melt flow index		18.1	17.9	17.9
	(9) Impact strength		30.2	30.4	31.1
	(10) Surface gloss characteristics		98.2	98.9	99
	(11) Gel evaluation		?	?	?
Thermo-plastic resin composition					

TABLE 10

			Comparative Examples			
Classification			22	23	24	25
Acrylic copolymer coagulant	Monomer mixture	MAA	5	5	6	6
		MA	—	—	—	—
		EA	95	95	94	94
		BA	—	—	—	—
	First emulsifier (parts by weight)		0.316	0.322	0.2	0.256
	Second emulsifier (parts by weight)		0.084	0.078	0.2	0.144
	n of Chemical Formula 1		1			
	X of Expression 1		5~7			
	Y of Expression 2		about 112.5~115.5		about 93.5~96.5	
	(1) Average particle diameter		116	128	80	92.5
Second diene-based rubber polymer	(2) Weight-average molecular weight		360,000	330,000	319,000	350,000
	(3) Polymerization conversion rate		>99.9	99.8	99.9	99.9
	Addition amount of coagulant (parts by weight)		1.2	1.2	1.2	1.2
	(4) Aggregation phenomenon		×	×	×	×
	(5) Average particle diameter		300	290	297	300
	(6) Particle size distribution		0.59	0.55	0.5	0.6
	(7) Amount of aggregate		120	350	390	400
	(8) Melt flow index		19	16.6	17.3	17
	(9) Impact strength		25	21.7	22.7	24.2
	(10) Surface gloss characteristics		97	90.2	91.6	96
Thermoplastic resin composition	(11) Gel evaluation		?	?	?	?

21

TABLE 11

Classification			Examples		
			14	15	16
Acrylic copolymer coagulant	Monomer mixture	MAA	6	6	6
		MA	—	—	—
		EA	94	94	94
		BA	—	—	—
	First emulsifier (parts by weight)		0.08	0.1	0.107
	Second emulsifier (parts by weight)		0.32	0.3	0.293
	n of Chemical Formula 1			1	
	X of Expression 1			5~7	
	Y of Expression 2			about 93.5~96.5	
	(1) Average particle diameter		93.5	95	96.5
(2) Weight-average molecular weight	303,000	308,000	298,000		
(3) Polymerization conversion rate		>99.9	99.8	99.9	

22

TABLE 11-continued

Classification		Examples		
		14	15	16
Second diene-based rubber polymer	Addition amount of coagulant (parts by weight)	1.2	1.2	1.2
	(4) Aggregation phenomenon	×	×	×
	(5) Average particle diameter	310	314	300
	(6) Particle size distribution	0.56	0.56	0.6
	(7) Amount of aggregate	100	130	190
	(8) Melt flow index	18	18.1	17.9
	(9) Impact strength	31	31.3	30.2
	(10) Surface gloss characteristics	98.1	98	98
	(11) Gel evaluation	?	?	?
Thermoplastic resin composition				

TABLE 12

Classification			Comparative Examples			
			26	27	28	29
Acrylic copolymer coagulant	Monomer mixture	MAA	6	6	7	7
		MA	—	—	—	—
		EA	94	94	93	93
		BA	—	—	—	—
	First emulsifier (parts by weight)		0.303	0.306	0.19	0.283
	Second emulsifier (parts by weight)		0.097	0.094	0.21	0.117
	n of Chemical Formula 1		1			
	X of Expression 1		5~7			
	Y of Expression 2		about 93.5~96.5		79.93~82.93	
	(1) Average particle diameter		97.5	99	71	79.5
	(2) Weight-average molecular weight	360,000	317,000	289,000	300,000	
	(3) Polymerization conversion rate	99.8	>99.9	>99.9	>99.9	
	Addition amount of coagulant (parts by weight)	1.2	1.2	1.2	1.2	
Second diene-based rubber polymer	(4) Aggregation phenomenon	×	×	×	×	
	(5) Average particle diameter	300	291	300	335	
	(6) Particle size distribution	0.62	0.59	0.45	0.44	
	(7) Amount of aggregate	500	420	500	600	
	(8) Melt flow index	17	16.9	18.9	18	
Thermoplastic resin composition	(9) Impact strength	24.9	23	18.6	25.7	
	(10) Surface gloss characteristics	95.1	96.7	97.2	98	
	(11) Gel evaluation	?	?	?	?	

23

TABLE 13

			Examples		
Classification			17	18	
Acrylic copolymer coagulant	Monomer mixture	MAA	7	7	5
		MA	—	—	
		EA	93	93	
		BA	—	—	
	First emulsifier (parts by weight)		0.312	0.313	10
	Second emulsifier (parts by weight)		0.088	0.087	
	n of Chemical Formula 1		1		15
	X of Expression 1		5~7		
	Y of Expression 2		79.93~82.93		20
	(1) Average particle diameter		80	82.5	
	(2) Weight-average molecular weight		299,000	297,000	25
Second diene-based rubber polymer	(3) Polymerization conversion rate		>99.9	99.8	
	Addition amount of coagulant (parts by weight)		1.2	1.2	30
	(4) Aggregation phenomenon		×	×	
	(5) Average particle diameter		314	320	25
	(6) Particle size distribution		0.47	0.48	
Thermoplastic resin composition	(7) Amount of aggregate		100	105	30
	(8) Melt flow index		18.9	18.5	
	(9) Impact strength		31.2	30.9	25
	(10) Surface gloss characteristics		97.9	98.1	
	(11) Gel evaluation		?	?	30

24

TABLE 14

			Comparative Examples			
Classification			30	31	32	
Acrylic coagulant	Monomer mixture	MAA	7	7	7.5	5
		MA	—	—	—	
		EA	93	93	92.5	
		BA	—	—	—	
	First emulsifier (parts by weight)		0.289	0.31	0.313	10
	Second emulsifier (parts by weight)		0.111	0.09	0.087	
	n of Chemical Formula 1			1		15
	X of Expression 1			5~7		
	Y of Expression 2		79.93~82.93		about 74.5~77.5	20
	(1) Average particle diameter		83.5	95	76	
	(2) Weight-average molecular weight		310,000	292,000	300,000	25
Second diene-based rubber polymer	(3) Polymerization conversion rate		99.8	99.9	99.9	
	Addition amount of coagulant (parts by weight)		1.2	1.2	1.2	30
	(4) Aggregation phenomenon		×	×	?	
	(5) Average particle diameter		348	351	—	25
	(6) Particle size distribution		0.46	0.6	—	
Thermoplastic resin composition	(7) Amount of aggregate		800	270	—	30
	(8) Melt flow index		17.2	18.1	—	
	(9) Impact strength		26	21.9	—	25
	(10) Surface gloss characteristics		98	99	—	
	(11) Gel evaluation		?	?	—	30

TABLE 15

			Comparative Examples					
Classification			33	34	35	36	37	38
Acrylic copolymer coagulant	Monomer mixture	MAA	16	16	16	18	18	18
		MA	—	—	—	—	—	—
		EA	—	—	—	—	—	—
		BA	84	84	84	82	82	82
	First emulsifier (parts by weight)		0.363	0.34	0.385	0.3	0.349	0.367
	Second emulsifier (parts by weight)		0.037	0.06	0.018	0.1	0.051	0.033
	n of Chemical Formula 1					3		
	X of Expression 1					16~18		
	Y of Expression 2		about 156.07~159.07				about 138.56~141.56	
	(1) Average particle diameter		157	131	180	110	140	160
	(2) Weight-average molecular weight		500,000	466,000	455,000	440,000	453,000	453,000
Second diene-based rubber polymer	(3) Polymerization conversion rate		>99.9	99.8	99.9	99.9	>99.9	99.8
	Addition amount of coagulant (parts by weight)		1.2	1.2	1.2	1.2	1.2	1.2
	(4) Aggregation phenomenon		×	○	×	×	×	×
	(5) Average particle diameter		306	—	344	321	331	348
	(6) Particle size distribution		0.66	—	0.68	0.59	0.64	0.7
	(7) Amount of aggregate		500	—	200	180	110	175

TABLE 15-continued

Classification		Comparative Examples					
		33	34	35	36	37	38
Thermoplastic resin composition	(8) Melt flow index	17.4	—	17	16.9	17.6	15.9
	(9) Impact strength	29	—	18.8	20.2	31.3	18.5
	(10) Surface gloss characteristics	90	—	92	90	84	84
	(11) Gel evaluation	Δ	—	×	×	Δ	×

TABLE 16

Classification		Comparative Examples		
		39	40	41
Acrylic copolymer coagulant	Monomer mixture	MAA	20	20
		MA	—	—
		EA	—	—
		BA	80	80
	First emulsifier (parts by weight)	0.3	0.342	0.358
	Second emulsifier (parts by weight)	0.1	0.058	0.042
	n of Chemical Formula 1		3	
	X of Expression 1		16~18	
	Y of Expression 2		about 124.55~127.55	
	(1) Average particle diameter	100	126	147
Second diene-based rubber polymer	(2) Weight-average molecular weight	411,000	408,000	427,000
	(3) Polymerization conversion rate	99.9	99.8	99.9
	Addition amount of coagulant (parts by weight)	1.2	1.2	1.2
	(4) Aggregation phenomenon	×	×	?
	(5) Average particle diameter	330	330	—
	(6) Particle size distribution	0.57	0.65	—
	(7) Amount of aggregate	400	102	—
	(8) Melt flow index	16	17.2	—
	(9) Impact strength	15.9	29.4	—
	(10) Surface gloss characteristics	88	84	—
Thermoplastic resin composition	(11) Gel evaluation	×	Δ	—

Referring to Table, it can be seen that Examples 1 to 10 exhibited a small amount of aggregates compared to Comparative Examples 1 to 17 and thus were excellent in latex stability, a melt flow index, impact strength, and surface gloss characteristics. It can be seen that Comparative Examples 1 and 2, in which the X value was below the range of Expression 1, exhibited degraded latex stability due to a large amount of aggregates and degraded processability due to a low melt flow index, and the impact strength and surface gloss characteristics thereof were also degraded. In addition, it can be seen that Comparative Example 2, in which the addition amount of a coagulant increased compared to Comparative Example 1, exhibited an improved melt flow index and improved impact strength, but surface gloss characteristics were degraded, and many gels were generated, leading to degraded appearance characteristics.

It can be seen that Comparative Examples 3 to 15, which did not satisfy Expression 2, exhibited degraded latex stability due to a large amount of aggregates and degraded processability due to a low melt flow index, and the impact strength and surface gloss characteristics thereof were also degraded.

In the case of Comparative Examples 16 and 17 in which the X value was above the range of Expression 1, an aggregation phenomenon occurred, and thus it was not possible to prepare a second diene-based rubber polymer.

It can be seen that Examples 11 to 16 exhibited a small amount of aggregates compared to Comparative Examples 18 to 32 and thus were excellent in latex stability, a melt flow index, impact strength, and surface gloss characteristics.

It can be seen that Comparative Examples 18 and 19, in which the X value was below the range of Expression 1, exhibited degraded latex stability due to a large amount of aggregates and degraded processability due to a low melt flow index, and the impact strength and surface gloss characteristics thereof were also degraded. In addition, it can be seen that Comparative Example 2, in which the addition amount of a coagulant increased compared to Comparative Example 1, exhibited an improved melt flow index and improved impact strength, but surface gloss characteristics were degraded, and many gels were generated, leading to degraded appearance characteristics.

It can be seen that Comparative Examples 20 to 31, which did not satisfy Expression 2, exhibited degraded latex stability due to a large amount of aggregates and degraded processability due to a low melt flow index, and the impact strength and surface gloss characteristics thereof were also degraded.

In the case of Comparative Example 32 in which Chemical Formula 1 and Expression 2 were satisfied, and the X value was above the range of Expression 1, an aggregation phenomenon occurred, and thus it was not possible to prepare a second diene-based rubber polymer.

It can be seen that Comparative Examples 33 and 37, which did not satisfy Chemical Formula 1, exhibited degraded latex stability due to a large amount of aggregates and degraded processability due to a low melt flow index, and the impact strength and surface gloss characteristics thereof were also degraded.

In the case of Comparative Example 34 in which Chemical Formula 1 was not satisfied, Expression 1 was satisfied, and the Y value was below the range of Expression 2, an aggregation phenomenon occurred, and thus it was not possible to prepare a second diene-based rubber polymer.

It can be seen that Comparative Examples 35 and 38, in which Chemical Formula 1 was not satisfied, Expression 1 was satisfied, and the Y value was above the range of Expression 2, exhibited degraded latex stability due to a large amount of aggregates and degraded processability due to a low melt flow index, and the impact strength and surface gloss characteristics thereof were also degraded, and an excessive amount of large-sized gels were generated, leading to poor appearance characteristics.

It can be seen that Comparative Example 36, in which Chemical Formula 1 was not satisfied, Expression 1 was satisfied, and the Y value was below the range of Expression 2, exhibited degraded latex stability due to a large amount of

27

aggregates and degraded processability due to a low melt flow index, and the impact strength and surface gloss characteristics thereof were also degraded, and an excessive amount of large-sized gels were generated, leading to poor appearance characteristics.

It can be seen that Comparative Example 39, in which Chemical Formula 1 and Expression 1 were not satisfied, and the Y value was below the range of Expression 2, exhibited degraded latex stability due to a large amount of aggregates and degraded processability due to a low melt flow index, and the impact strength and surface gloss characteristics thereof were also degraded, and an excessive amount of large-sized gels were generated, leading to poor appearance characteristics.

It can be seen that Comparative Example 40, which did not satisfy Chemical Formula 1 and Expression 1 and satisfied Expression 2, exhibited degraded latex stability due to a large amount of aggregates and degraded processability due to a low melt flow index, and the impact strength and surface gloss characteristics thereof were also degraded, and an excessive amount of large-sized gels were generated, leading to poor appearance characteristics.

In the case of Comparative Example 41 in which Chemical Formula 1 and Expression 1 were not satisfied, and the Y value was above the range of Expression 2, an aggregation phenomenon occurred, and thus it was not possible to prepare a second diene-based rubber polymer.

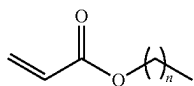
The invention claimed is:

1. An acrylic copolymer coagulant which is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and an acrylic monomer represented by Chemical Formula 1 below,

wherein a content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant satisfies Expression 1 below, and

an average particle diameter Y of the acrylic copolymer coagulant satisfies Expression 2 below,

wherein the acrylic copolymer coagulant has a weight-average molecular weight of 700,000 g/mol to 950,000 g/mol:



<Chemical Formula 1>

in Chemical Formula 1,
n is 0,

$$2+(n+1)^2-0.5(n+1)\leq X\leq 2+(n+1)^2+0.5(n+1) \quad \text{<Expression 1>}$$

$$\frac{\{20.817\times(n+1)^2-27.35\times(n+1)+28.433\}}{X\times 10-1.5}\leq Y\leq \frac{\{20.817\times(n+1)^2-27.35\times(n+1)+28.433\}}{X\times 10+1.5} \quad \text{<Expression 2>}$$

28

in Expressions 1 and 2,

n is an n value of Chemical Formula 1,

X is a content of a carboxylic acid-based monomer unit comprised in the acrylic copolymer coagulant (units: wt %), and

Y is an average particle diameter of the acrylic copolymer coagulant (units: nm).

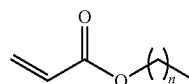
2. The acrylic copolymer coagulant of claim 1, wherein an average particle diameter Y of the acrylic copolymer coagulant is in a range of 61.5 nm to 89.1 nm.

3. An acrylic copolymer coagulant which is a copolymer of a monomer mixture comprising a carboxylic acid-based monomer and an acrylic monomer represented by Chemical Formula 1 below,

wherein a content X of a carboxylic acid-based monomer unit in the acrylic copolymer coagulant satisfies Expression 1 below, and

an average particle diameter Y of the acrylic copolymer coagulant satisfies Expression 2 below,

wherein the acrylic copolymer coagulant has a weight-average molecular weight of 250,000 g/mol to 350,000 g/mol:



<Chemical Formula 1>

in Chemical Formula 1,

n is 1,

$$2+(n+1)^2-0.5(n+1)\leq X\leq 2+(n+1)^2+0.5(n+1) \quad \text{<Expression 1>}$$

$$\frac{\{20.817\times(n+1)^2-27.35\times(n+1)+28.433\}}{X\times 10-1.5}\leq Y\leq \frac{\{20.817\times(n+1)^2-27.35\times(n+1)+28.433\}}{X\times 10+1.5} \quad \text{<Expression 2>}$$

in Expressions 1 and 2,

n is an n value of Chemical Formula 1,

X is a content of a carboxylic acid-based monomer unit comprised in the acrylic copolymer coagulant (units: wt %), and

Y is an average particle diameter of the acrylic copolymer coagulant (units: nm).

4. The acrylic copolymer coagulant of claim 3, wherein an average particle diameter Y of the acrylic copolymer coagulant is in a range of 80 nm to 115.5 nm.

5. A method of preparing a graft copolymer, comprising: preparing a second diene-based rubber polymer by coagulation of a first diene-based rubber polymer with the acrylic copolymer coagulant according to claim 1; and preparing a graft copolymer by graft polymerization of the second diene-based rubber polymer with an aromatic vinyl-based monomer and a vinyl cyanide-based monomer.

6. The method of claim 5, wherein the first diene-based rubber polymer has an average particle diameter of 90 to 110 nm, and the second diene-based rubber polymer has an average particle diameter of 250 to 350 nm.

* * * * *